higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

MATHEMATICS N5

(16030175)

31 March 2023 (X-paper)
09:00–12:00

This question paper consists of 7 pages and a formula sheet of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MATHEMATICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only a blue or black pen.
5. Show all intermediate steps and simplify where possible.
6. All final answers must be rounded off to THREE decimal places.
7. Questions may be answered in any order, but subsections of questions must be kept together.
8. Sketches must be large, neat, and fully labelled.
9. Scientific calculators may be used.
10. Write neatly and legibly.

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QUESTION 1

1.1 Determine the following limit:

$$y = \lim_{x \to 3} \left[\frac{1}{\ln(x-2)} - \frac{1}{(x-3)} \right]$$ (4)

1.2 Determine the value(s) of x for which $f(x)$ is discontinuous if

$$y = \frac{x^3 + 5x - 10}{x^2 - 3x + 2}$$ (2)

[6]

QUESTION 2

2.1 Determine $\frac{dy}{dx}$ using first principles:

$$y = \frac{x - 10}{x + 2}$$ (4)

2.2 Determine $\frac{dy}{dx}$ in each of the following cases:
(Simplification NOT required)

2.2.1
$$y = \sin^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$$ (4)

2.2.2
$$y = \frac{e^{x \cos x}}{1 + e^x}$$ (4)

2.3 Calculate $\frac{dy}{dx}$ if $y = (3x - 2^x)^{\cos 3x}$ with aid of logarithmic differentiation. (4)

2.4 Given the implicit function:

$$\frac{9 - x^2}{(y - x)^3} = 1$$

2.4.1 Determine $\frac{dy}{dx}$ (3)

2.4.2 Determine the equation of the tangent to the graph at the point $(1, 3)$. (3)

[22]

QUESTION 3

3.1 Given: $f(x) = x^3 - 2x^2 - 3x + 5$

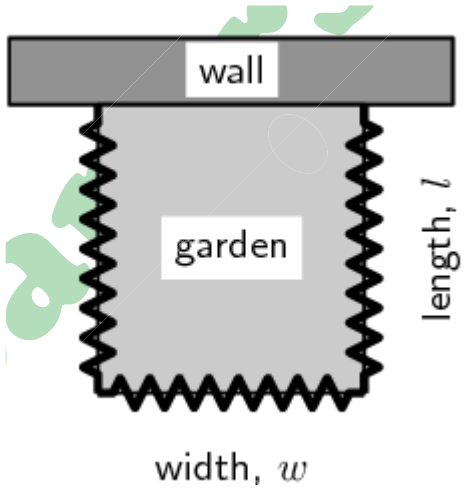
3.1.1 Determine the coordinates of the turning points of $f(x)$. (3)

3.1.2 Draw up a table of x and $f(x)$, where x is ranging from $x = -2$ to $x = 3$. (2)

3.1.3 Draw a neat graph of $f(x)$ between these values. Show the turning points on it. (2)

3.1.4 One root of the equation $x^3 - 2x^2 - 3x + 5$ is close to 1,3. Use Taylor's/ Newton's method TWICE to determine a better approximation of this root (root to be correct to THREE decimal figures). (4)

3.2 Michael wants to start a vegetable garden. He decides to fence it off in the shape of a rectangle from the rest of the garden. Michael has only 160 m of fencing, so he decides to use a wall as one border of the vegetable garden. Calculate the width and length of the garden that corresponds to the largest possible area that Michael can fence off.



(5)

3.3 You are inflating a spherical balloon at the rate of 7 cm^3/s. How fast is its radius increasing when the radius is 4 cm?

HINT: $V = \frac{4}{3}\pi r^2$

(4)

[20]

QUESTION 4

4.1 Determine $\int y\,dx$ in each of the following cases:

4.1.1 $$y = \frac{e^{\sqrt{\sin x}}}{\sec x.\sqrt{\sin x}}$$ (3)

4.1.2 $$y = \sin 3x \sin(-2x)$$ (3)

4.1.3 $$y = \cos^2 x \sin^3 x$$ (5)

4.1.4 $$y = \frac{3x^3 - 2x^2 + x - 5}{x + 2}$$ (5)

4.1.5 $$y = \frac{x^3 e^{x^2}}{(x^2+1)^2}$$

HINT: $\text{let } u \text{ or } f(x) = x^2 e^{x^2}$ (4)

4.2 Determine $\int y\,dx$ by resolving the integral into partial fractions:

$$y = \frac{11 - 3x}{x^2 + 2x - 3}$$ (5)

[25]

QUESTION 5

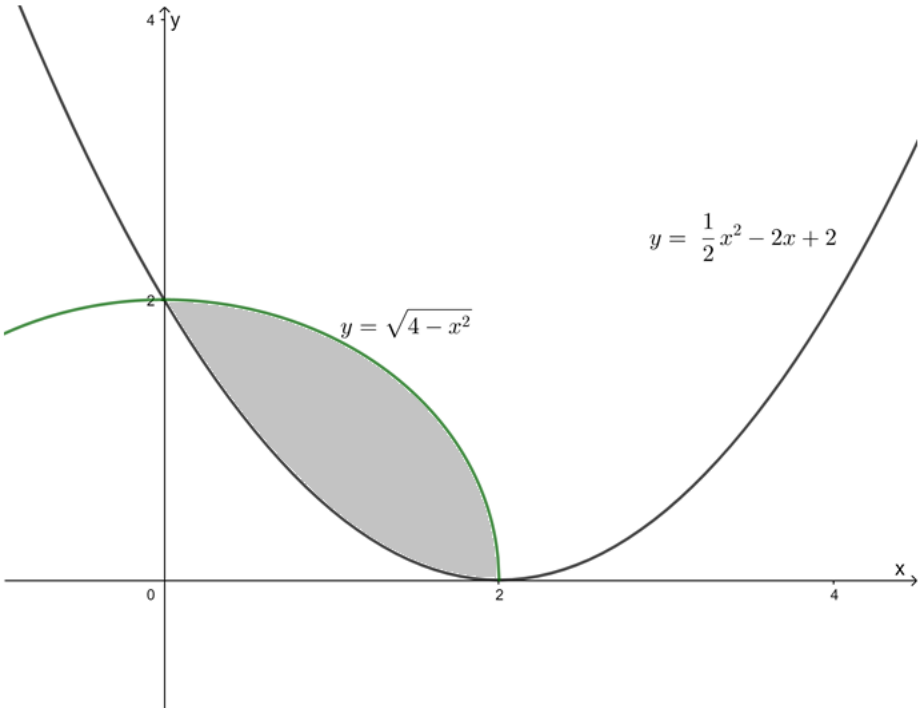
5.1 Evaluate the following definite integral:

$$\int_0^{\frac{\pi}{4}} \frac{\sec x \tan x}{1+\sec^2 x}\,dx$$ (5)

5.2 State the value of the Laplace transform L(3). (1)

5.3 The graphs of the following functions are given below:

$y = \sqrt{4-x^2}$ and $y = \frac{1}{2}x^2 - 2x + 2$



5.3.1 Calculate the magnitude of the area enclosed by the graphs. (3)

5.3.2 Calculate the volume of the solid of revolution formed when the area in QUESTION 5.3.1 above is rotated about the x-axis. (5)

5.4 Calculate the moment of inertia of a circular lamina with a radius of 4 cm and a mass of 25g about an axis through its centre and perpendicular to the plane of the lamina. (5)

[19]

QUESTION 6

6.1 Determine the particular solution of:

$$\frac{dy}{dx} = 2x - 1 + 2xy - y \text{ , if } y(0) = 2 \qquad (3)$$

6.2 Determine the particular solution of:

$$\frac{1}{(x+3)^2 - 2}\frac{d^2y}{dx^2} = 1, \quad \text{if} \quad \frac{dy}{dx}_{|\ x=1} = 0 \text{ and } y(1) = 2 \qquad (5)$$

[8]

TOTAL: 100

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MATHEMATICS N5

FORMULA SHEET

Any applicable formula may also be used.

TRIGONOMETRY

$\sin^2 x + \cos^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

$1 + \cot^2 x = \operatorname{cosec}^2 x$

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$\sin^2 A = ½ - ½ \cos 2A$

$\cos^2 A = ½ + ½ \cos 2A$

$\sin (A \pm B) = \sin A \cos B \pm \sin B \cos A$

$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$

$$\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$\sin A \cos B = ½ [\sin (A + B) + \sin (A - B)]$

$\cos A \sin B = ½ [\sin (A + B) - \sin (A - B)]$

$\cos A \cos B = ½ [\cos (A + B) + \cos (A - B)]$

$\sin A \sin B = ½ [\cos (A - B) - \cos (A + B)]$

$$\tan x = \frac{\sin x}{\cos x} \; ; \; \sin x = \frac{1}{\operatorname{cosec} x} \; ; \; \cos x = \frac{1}{\sec x}$$

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BINOMIAL THEOREM

$$(x+h)^n = x^n + nx^{n-1}\,h + \frac{n(n-1)}{2!}x^{n-2}\,h^2 + \ldots$$

DIFFERENTIATION

$$e = -\frac{f(a)}{f'(a)}$$

$$r = a + e$$

PRODUCT RULE

$$y = u(x)\,.\,v(x)$$

$$\frac{dy}{dx} = u\,.\,\frac{dv}{dx} + v\,.\,\frac{du}{dx}$$

$$= u \cdot v' + v \cdot u'$$

QUOTIENT RULE

$$y = \frac{u(x)}{v(x)}$$

$$\frac{dy}{dx} = \frac{v\,.\,\frac{du}{dx} - u\,.\,\frac{dv}{dx}}{v^2}$$

$$= \frac{v \cdot u' - u \cdot v'}{v^2}$$

CHAIN RULE

$$y = f(u(x))$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

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$f(x)$	$\frac{d}{dx} f(x)$	$\int f(x)dx$
ax^n	nax^{n-1}	$\frac{ax^{n+1}}{n+1} + c$
a	0	$ax + c$
e^x	e^x	$e^x + c$
a^x	$a^x.\ln a$	$\frac{a^x}{\ln a} + c$
$\log_e x$	$\frac{1}{x}$	—
$\log_a x$	$\frac{1}{x \ln a}$	—
$\sin x$	$\cos x$	$-\cos x + c$
$\cos x$	$-\sin x$	$\sin x + c$
$\tan x$	$\sec^2 x$	$\ln(\sec x) + c$
$\cot x$	$-\text{cosec}^2 x$	$\ln(\sin x) + c$
$\sec x$	$\sec x \cdot \tan x$	$\ln[\sec x + \tan x] + c$
$\text{cosec } x$	$-\text{cosec } x \cdot \cot x$	$\ln(\text{cosec} x - \cot x) + c$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	—
$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$	—
$\tan^{-1} x$	$\frac{1}{1+x^2}$	—
$\cot^{-1} x$	$\frac{-1}{1+x^2}$	—
$\sec^{-1} x$	$\frac{1}{x\sqrt{x^2-1}}$	—
$\text{cosec}^{-1} x$	$\frac{-1}{x\sqrt{x^2-1}}$	—

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$f(x)$	$\frac{d}{dx} f(x)$	$\int f(x)dx$
$\frac{1}{\sqrt{a^2 - x^2}}$	—	$\sin^{-1}\left(\frac{x}{a}\right) + c$
$\frac{1}{a^2 + x^2}$	—	$\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$
$\frac{1}{x\sqrt{x^2 - a^2}}$	—	$\frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + c$
$\sqrt{a^2 - x^2}$	—	$\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + \frac{x}{2}\sqrt{a^2 - x^2} + c$
$\frac{1}{x^2 - a^2}$	—	$\frac{1}{2a} ln\left(\frac{x - a}{x + a}\right) + c$
$\frac{1}{a^2 - x^2}$	—	$\frac{1}{2a} ln\left(\frac{a + x}{a - x}\right) + c$

INTEGRATION

$\int f(x)\, g'(x)\, dx = f(x)\, g(x) - \int f'(x)\, g(x)\, dx$

$\int \frac{f'(x)}{f(x)} dx = \ln f(x) + c$

$\int [f(x)]^n f'(x)\, dx = \frac{[f(x)]^{n+1}}{n+1} + c$

$\frac{f(x)}{(ax+b)(cx+d)} = \frac{A}{ax+b} + \frac{B}{cx+d}$

$\frac{f(x)}{(x+a)^n} = \frac{A}{(x+a)} + \frac{B}{(x+a)^2} + \frac{C}{(x+a)^3} + \cdots \frac{Z}{(x+a)^n}$

APPLICATIONS OF INTEGRATION

AREAS

$A_x = \int_a^b y dx \; ; \; A_x = \int_a^b (y_1 - y_2)\, dx$

$A_y = \int_a^b x dy \; ; \; A_y = \int_a^b (x_1 - x_2)\, dy$

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VOLUMES

$$V_x = \pi \int_a^b y^2 dx \; ; \; V_x = \pi \int_a^b \left(y_1^2 - y_2^2 \right) dx$$

$$V_y = \pi \int_a^b x^2 dy \; ; \; V_y = \pi \int_a^b \left(x_1^2 - x_2^2 \right) dy$$

SECOND MOMENT OF AREA

$$I_x = \int_a^b r^2 dA \; ; \; I_y = \int_a^b r^2 dA$$

MOMENTS OF INERTIA

Mass = density × volume

$M = \rho V$

DEFINITION: $I = m r^2$

GENERAL: $I = \int_a^b r^2 dm = \rho \int_a^b r^2 dV$

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